Trainings

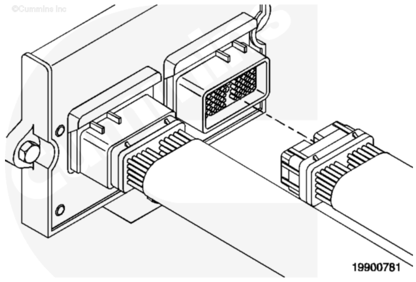
Resistance Check

The engine position sensor circuit includes pin 10 (+5-VDC supply wire), pin 9 (signal wire), and pin 19 (sensor return).

Disconnect the engine harness from the ECM.

Disconnect the engine harness from the engine position sensor.

Check for damaged pins.

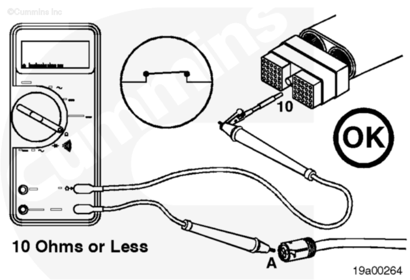


+5-VDC Supply Wire Resistance - Checking

⚠ CAUTION ⚠

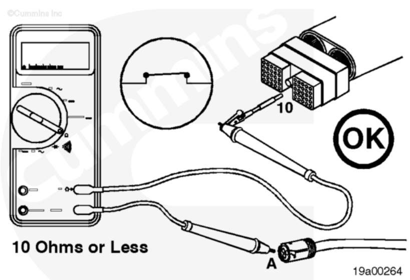
Do not use probes or leads other than Part Number 3822758. The connector will be damaged. The leads must fit tightly in the connector without expanding the pins in the connector.

Insert the test lead into pin 10 of the engine harness connector. Connect the alligator clip to the multimeter probe. Touch the other multimeter probe to pin A of the engine position sensor connector, harness side.



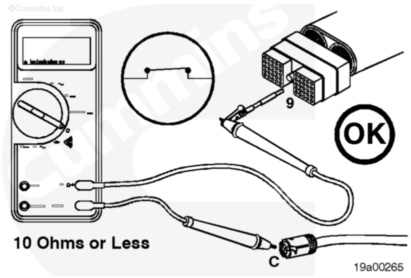
Measure the resistance. The multimeter **must** show a closed circuit (10 ohms or less). If more than 10 ohms are measured, there is an open circuit in the +5-VDC supply wire. Repair the +5-VDC supply wire or replace the engine harness.

Refer to Procedure 019-250 (/qs3/pubsys2/xml/en/procedures/87/87-019-250.html), Refer to Procedure 019-203 (/qs3/pubsys2/xml/en/procedures/99/99-019-203.html), or Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/87/87-019-043.html).

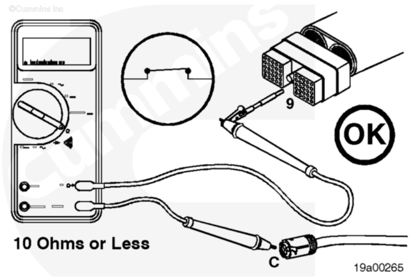


Signal Wire Resistance - Checking

Insert the test lead into pin 9 of the engine harness connector. Connect the alligator clip to the multimeter probe. Touch the other probe to pin C of the engine position sensor connector, harness side.



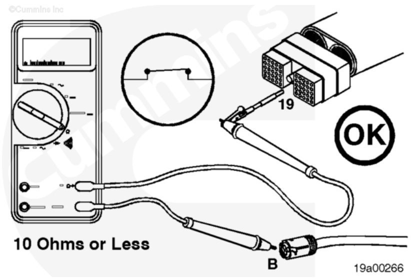
Measure the resistance. The multimeter **must** show a closed circuit (10 ohms or less). If more than 10 ohms are measured, there is an open circuit in the signal wire. Repair the signal wire or replace the engine harness.



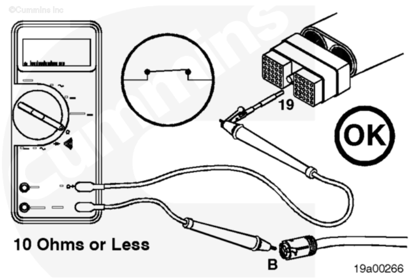
Refer to Procedure 019-250 (/qs3/pubsys2/xml/en/procedures/87/87-019-250.html), Refer to Procedure 019-203 (/qs3/pubsys2/xml/en/procedures/99/99-019-203.html), or Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/87/87-019-043.html).

Return Wire Resistance - Checking

Insert the test lead into pin 19 of the engine harness connector. Connect the alligator clip to the multimeter probe. Touch the other probe to pin B of the engine position sensor connector, harness side.



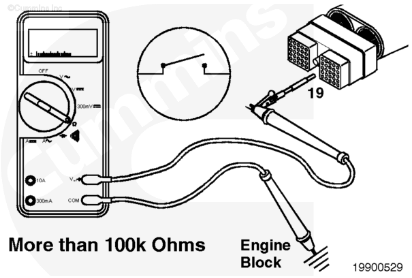
Measure the resistance. The multimeter **must** show a closed circuit (10 ohms or less). If more than 10 ohms are measured, there is an open circuit in the return wire. Repair the return wire or replace the engine harness.



Refer to Procedure 019-250 (/qs3/pubsys2/xml/en/procedures/87/87-019-250.html), Refer to Procedure 019-203 (/qs3/pubsys2/xml/en/procedures/99/99-019-203.html), or Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/87/87-019-043.html).

Check for Short Circuit to Ground

Return Wire - Checking

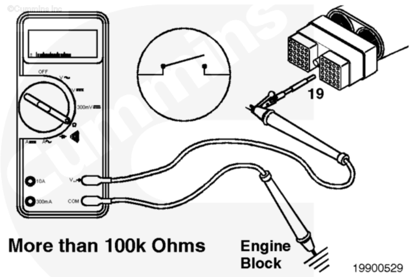


⚠ CAUTION ⚠

Do not use probes or test leads other than Part Number 3822758. The connector will be damaged. The leads must fit tightly in the connector without expanding the pins in the connector.

Disconnect the engine harness from the ECM.

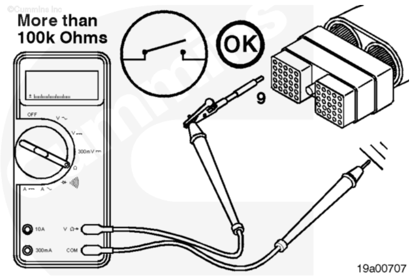
Insert the test lead into pin 19 of the engine harness connector. Connect the alligator clip to the multimeter probe. Touch the other probe to the engine block.



Measure the resistance. The multimeter **must** show an open circuit (100k ohms or more). If less than 100k ohms are measured, there is a short circuit to ground in the return wire.

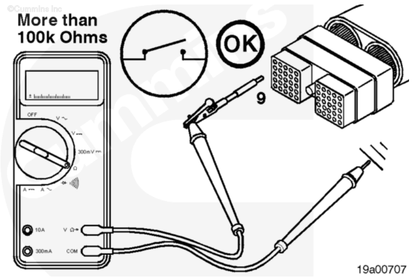
Repair the return wire or replace the engine harness. Refer to Procedure 019-250 (/qs3/pubsys2/xml/en/procedures/87/87-019-250.html), Refer to Procedure 019-203 (/qs3/pubsys2/xml/en/procedures/99/99-019-203.html), or Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/87/87-019-043.html).

Signal Wire - Checking



Make sure the engine position sensor is disconnected from the engine harness.

Insert the test lead into pin 9 of the engine harness connector. Connect the alligator clip to the multimeter probe. Touch the other probe to the engine block.

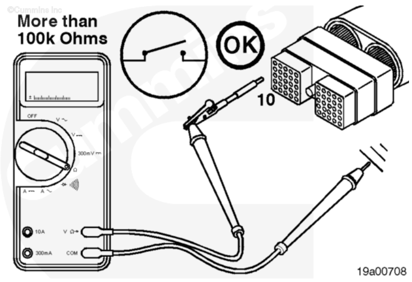


Measure the resistance. The multimeter **must** show an open circuit (100k ohms or more). If less than 100k ohms are measured, there is a short circuit to ground in the signal wire. Repair the signal wire or replace the engine harness.

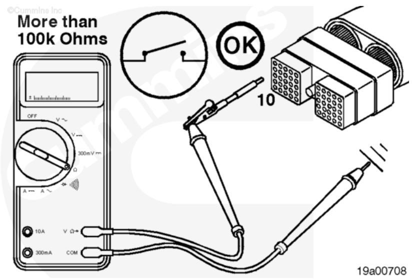
Refer to Procedure 019-250 (/qs3/pubsys2/xml/en/procedures/87/87-019-250.html), Refer to Procedure 019-203 (/qs3/pubsys2/xml/en/procedures/99/99-019-203.html), or Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/87/87-019-043.html).

+5-VDC Supply Wire - Checking

Insert the test lead into pin 10 of the engine harness connector. Connect the alligator clip to the multimeter probe. Touch the other probe to the engine block.



Measure the resistance. The multimeter **must** show an open circuit (100k ohms or more). If less than 100k ohms are measured, there is a short circuit to ground in the +5-VDC supply wire.



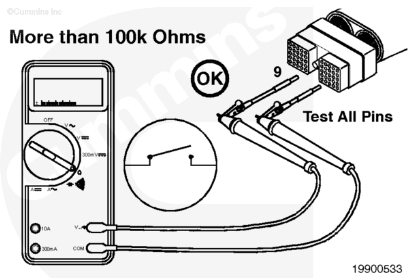
Repair the +5-VDC supply wire or replace the engine harness.

Refer to Procedure 019-250 (/qs3/pubsys2/xml/en/procedures/87/87-019-250.html), Refer to Procedure 019-203 (/qs3/pubsys2/xml/en/procedures/99/99-019-203.html), or Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/87/87-019-043.html).

Check for Short Circuit from Pin-to-Pin

⚠ CAUTION ⚠

Do not use probes or test leads other than Part Number 3822758. The connector will be damaged. The leads must fit tightly in the connector without expanding the connector pins.



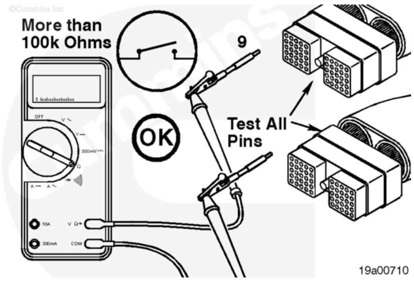
Signal Wire - Checking

Disconnect the OEM and engine harnesses from the ECM.

Insert a test lead into pin 9 of the engine harness connector. Connect the alligator clips to the multimeter probes. Insert the other lead into all other pins of the engine harness connector. Measure the resistance.

Then, repeat the pin-to-pin check from pin 9 of the engine harness connector to all pins in the OEM interface harness connector.

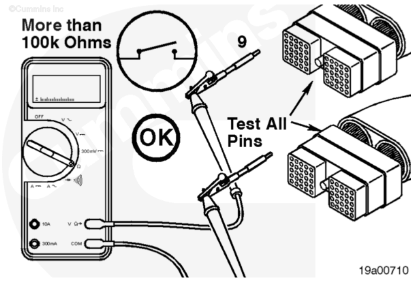
Measure the resistance.



For all pin checks, the multimeter **must** show an open circuit (100k ohms or more). If the circuit is **not** open, there is a short circuit between the signal wire pin and any pin that measured a closed circuit.

Repair or replace the engine harness.

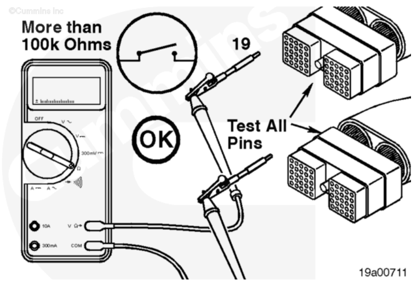
Refer to Procedure 019-250 (/qs3/pubsys2/xml/en/procedures/87/87-019-250.html) or Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/87/87-019-043.html).



Return Wire - Checking

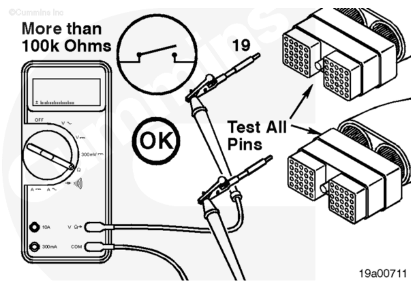
Insert a test lead into pin 19 of the engine harness connector. Connect the alligator clips to the multimeter probes. Insert the other lead into all other pins of the engine harness connector.

Measure the resistance.



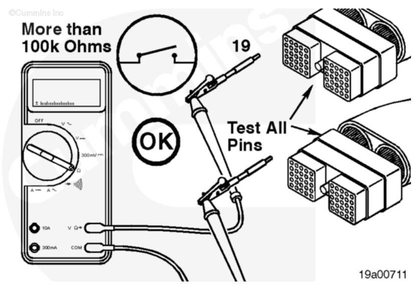
Then, repeat the pin-to-pin check from pin 19 of the engine harness connector to all pins in the OEM interface harness connector.

Measure the resistance.



For all pin checks, the multimeter **must** show an open circuit (100k ohms or more). If the circuit is **not** open, there is a short circuit between the return wire pin and any pin that measured a closed circuit. Repair or replace the engine harness.

Refer to Procedure 019-250 (/qs3/pubsys2/xml/en/procedures/87/87-019-250.html) or Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/87/87-019-043.html).



Last Modified: 09-Aug-2018